Do Now · retrieval

Specification gap and misconception check

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does a salt add to password storage?

Q2

What is an INNER JOIN result based on?

Q3

Which OS mechanism translates pages to frames?

Q4

Full adder A=1,B=0,carry-in=0: sum and carry, and binary total.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — When correcting an error, identify the incorrect assumption precisely; a bare True/False is weak diagnostic evidence.

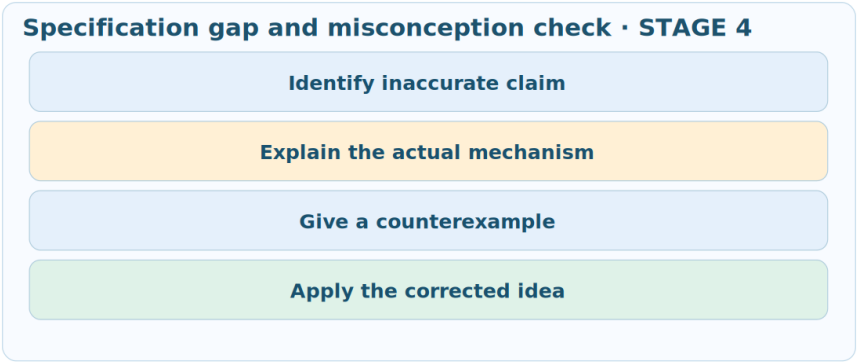
Core idea 2

Trace — Connect each correction to a worked example: HTTPS versus phishing, paging versus remote files, key joins versus unmatched rows.

Core idea 3

Evaluate — Show transfer by applying the rule to a different problem without being prompted by the original misconception.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

misconception — Incorrect mental model of a mechanism.

counterexample — Case disproving a general claim.

transfer — Apply understanding to new case.

diagnostic — Detect specific gaps in knowledge.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Correct 'a page fault is a missing website page'.

Follow the worked method

1. Predict first — A page fault is a memory-management event where required virtual page data is not currently resident.

2. Show method — HTTP 404 is a web server status about a requested resource; the word 'page' refers to different concepts.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

The input has no repeated runs longer than one symbol.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Correct four statements: 'DNS sends HTML'; 'ROM is volatile'; 'SHA is encryption'; 'INNER JOIN keeps all rows'. [8]

Question 2

Explain why a school's cloud files and its RAM paging system should not be described by one mechanism. [6]

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Choose three of the previous misconceptions and construct a counterexample for each. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

An accurate definition alone guarantees that you can apply a concept.

Use key vocabulary from memory

1. Explain misconception in your own words.

2. Explain counterexample in your own words.

3. Explain transfer in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.